Consorzio Interuniversitario Nazionale per la Scienza e Tecnologia dei Materiali
Italy

November 5, 2026

Editor-in-Chief
International Journal of Quantum Chemistry

Dear Editor,

I am pleased to submit the manuscript entitled *"Semiempirical Orbital-Optimized DFT in MOUSE: Spin-Pure Excited-State Surfaces with Analytic Gradients and Hessians"* for consideration as a Full Paper in the *International Journal of Quantum Chemistry*.

The manuscript addresses a practical limitation of excited-state quantum chemistry: repeated calculations on large molecules require a state-specific energy surface with reliable gradients and curvatures, while conventional response methods are primarily designed for efficient vertical multi-root spectra. We develop a DFT-informed semiempirical orbital-optimized model in MOUSE that combines a one-time DFT reference with a spin-polarized GFN2 valence calculation, spin-pure singlet configurations, selected screened double excitations, and complete analytic response through the nuclear Hessian.

The principal results are: (i) a common-orbital spin-pure excited-state surface with explicit state following and no fitted excitation-energy offset; (ii) analytic gradients and Hessians including orbital stationarity, spin recoupling, configuration-space response, the moving ground projector and state-eigenvector response; (iii) a 26-state development assessment with 0.183 eV mean absolute error, including a marked improvement for the $\pi \rightarrow \pi^*$ subset over a matched TDA control; and (iv) larger carotene and chlorophyllide examples demonstrating configuration-space tracking and convergence diagnostics. The manuscript is positioned relative to simplified TD-DFT/sTDA, TD-DFTB, ΔDFTB, TDDFT+TB and semiempirical multireference alternatives, while clearly separating demonstrated results from future independent-transferability tests.

This work is appropriate for IJQC because it presents a quantum-chemical methodological development with an explicit variational formulation, analytic nuclear response and an implementation intended for molecular simulation and multilevel QM/QM$'$ applications. The accompanying Supporting Information contains the full protocol, provenance, machine-readable tables, derivative checks and additional convergence analyses.

The manuscript is original, has not been published in full, and is not under consideration elsewhere. The author declares no conflict of interest. All data and source code required to reproduce the reported results are available in the MOUSE repository; large archived checkpoints are retained with checksummed provenance and will be deposited in a permanent public archive before publication.

Suggested reviewers are experts in excited-state electronic structure, orbital-optimized DFT and semiempirical quantum chemistry. We would be happy to provide specific names and contact details through the submission system. We have no non-preferred reviewer requests.

Thank you for considering this manuscript. I look forward to your response.

Sincerely,

Vincenzo Barone